

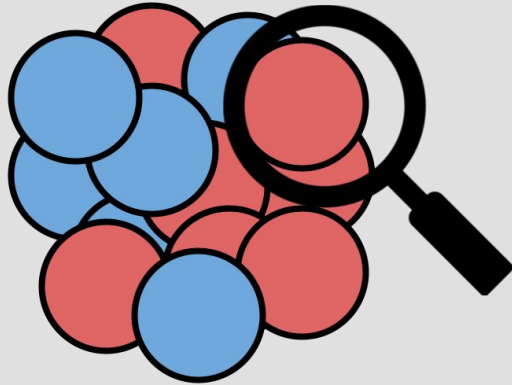
The Role of Single Pion Exchange in the Binding of the Omega Meson

...

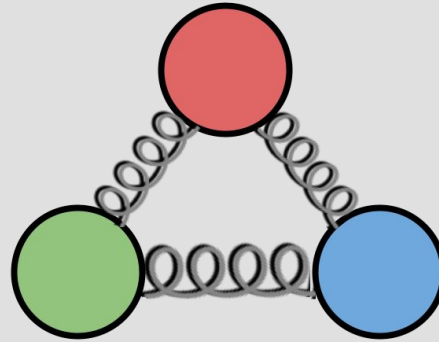
Connor Cassidy
Advisor: Dr. Andrew Jackura

Quarks, Gluons, and QCD

- Quarks: fundamental particles, six flavors, interact under strong force
- Gluons: mediators of the strong interaction
- Quantum Chromodynamics (QCD): theory of quark-gluon interactions
- Hadrons: composite particles made of multiple quarks



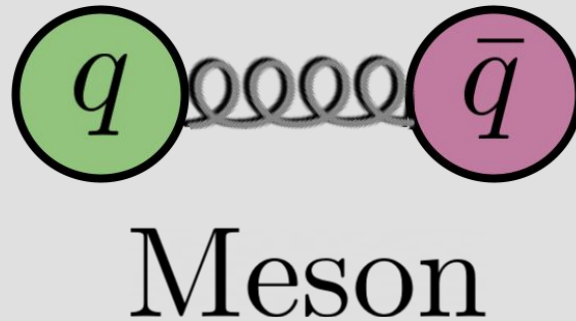
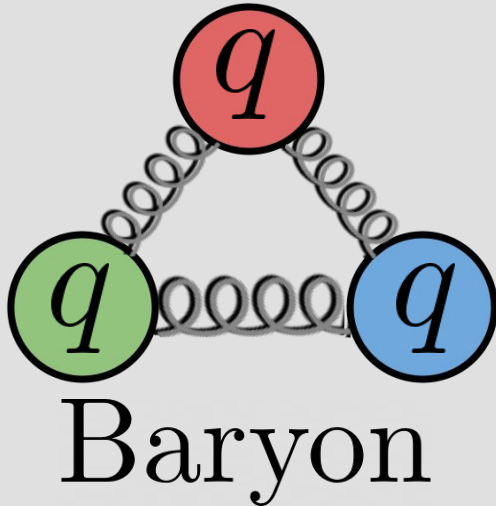
Nucleus



Hadron

Hadrons

- Conserved quantum numbers (spin, parity, isospin, etc.)
- Baryons: hadrons with half-integer spins, three valence quarks
- Mesons: hadrons with integer spin, quark-antiquark pairs
- Hadrons come in “families,” with excited states



Examples of Hadrons

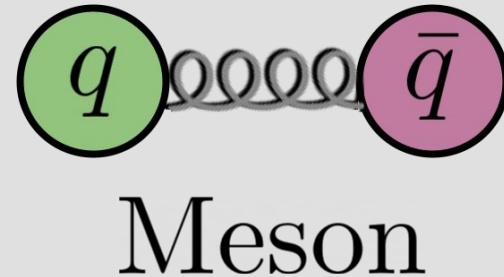
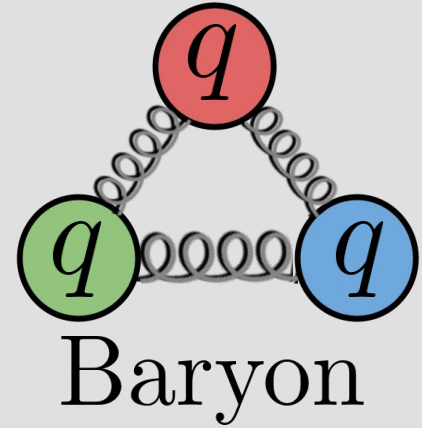
$$I(J^P)$$

Baryons:

- Nucleons (Protons and Neutrons) $\frac{1}{2} \left(\frac{1}{2}^+ \right)$

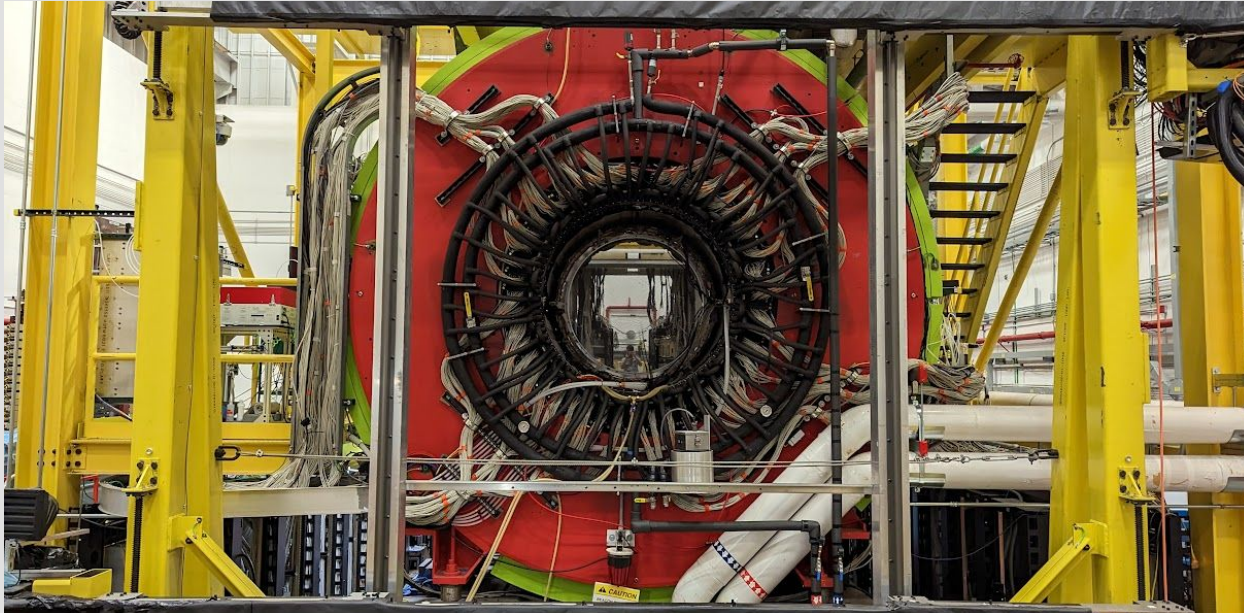
Mesons

- Pion (π) $1(0^-)$
- Rho Meson (ρ) $1(1^-)$
- Omega Meson (ω) $0(1^-)$



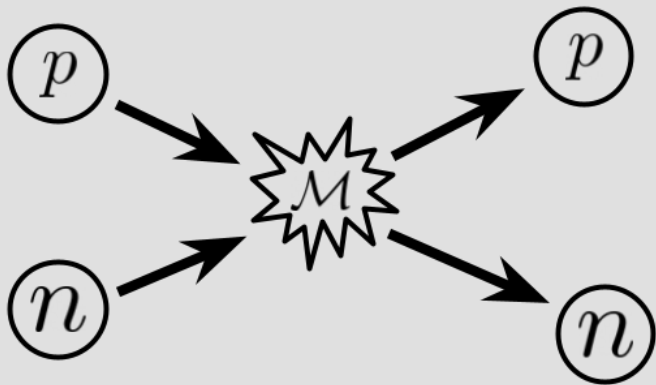
Hadron Spectroscopy

- QCD difficult to calculate directly, can learn by studying patterns in hadron properties
- Study spectrum of hadron masses
- Studied experimentally at facilities such as JLab



Scattering Theory

- Study hadrons through their reactions
- Cross section: quantity related to the likelihood of scattering occurring between particles, experimentally measured
- Scattering amplitude: function of kinematic variables, related to cross section, useful for theoretical calculations
- Example: proton and neutron scattering...

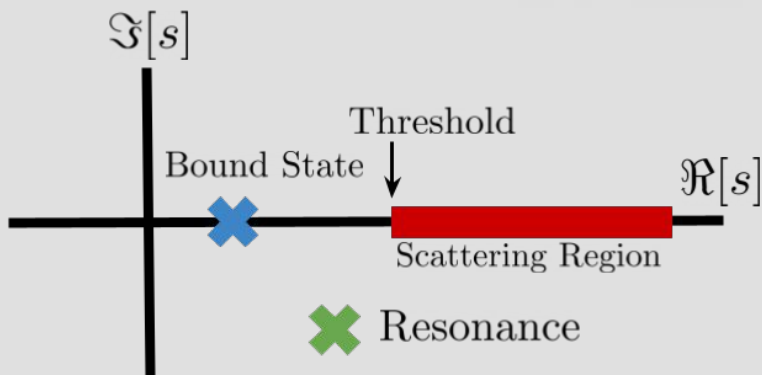
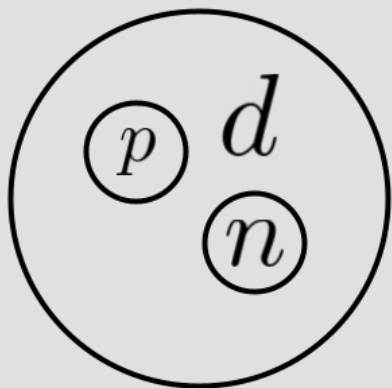


$$\sigma \propto |\mathcal{M}|^2$$

Bound States and Resonances

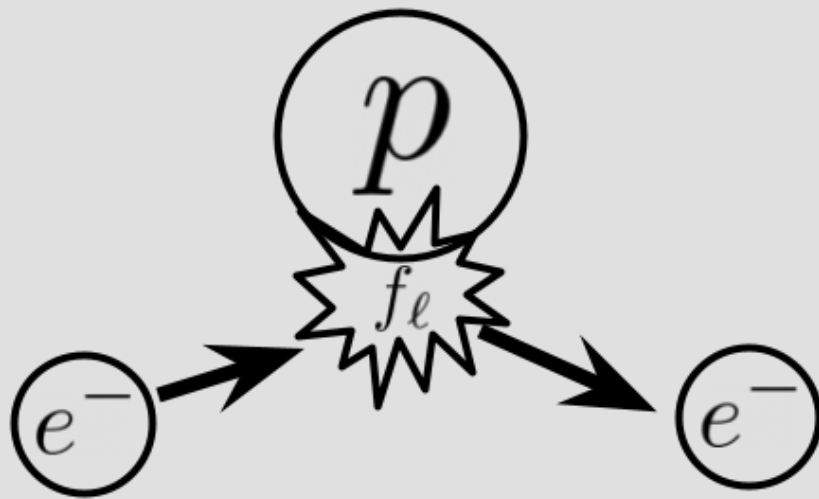
- Ex. proton-neutron scattering can form a deuteron, simplest nucleus
- Threshold: minimum scattering energy of system
- Scattering amplitude can be continued below threshold to find poles
- Two locations for poles:
 - Real axis below threshold - bound state (stable)
 - Elsewhere off the real axis - resonance (decays)

$$\mathcal{M} \sim \frac{-g_b^2}{s - s_b}$$



An Example: Non-Relativistic Electron-Proton Scattering

- Normally in QM, energy levels of hydrogen are found through solving Schrödinger equation
- Alternative: solve for the electron proton scattering amplitude for definite angular momentum ℓ



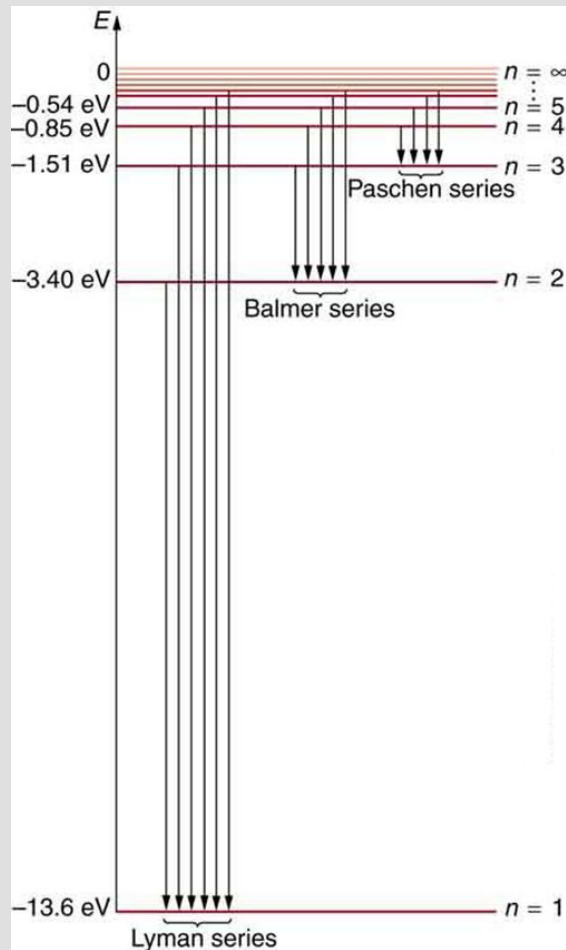
$$f_\ell \sim \Gamma \left(\ell + 1 + i \frac{\mu \alpha}{k} \right)$$

The Hydrogen Atom

- Gamma function has poles at non-positive integers
- Poles show up at the exact energy levels of the Hydrogen atom
- Constraint of $\ell < n$ appears naturally

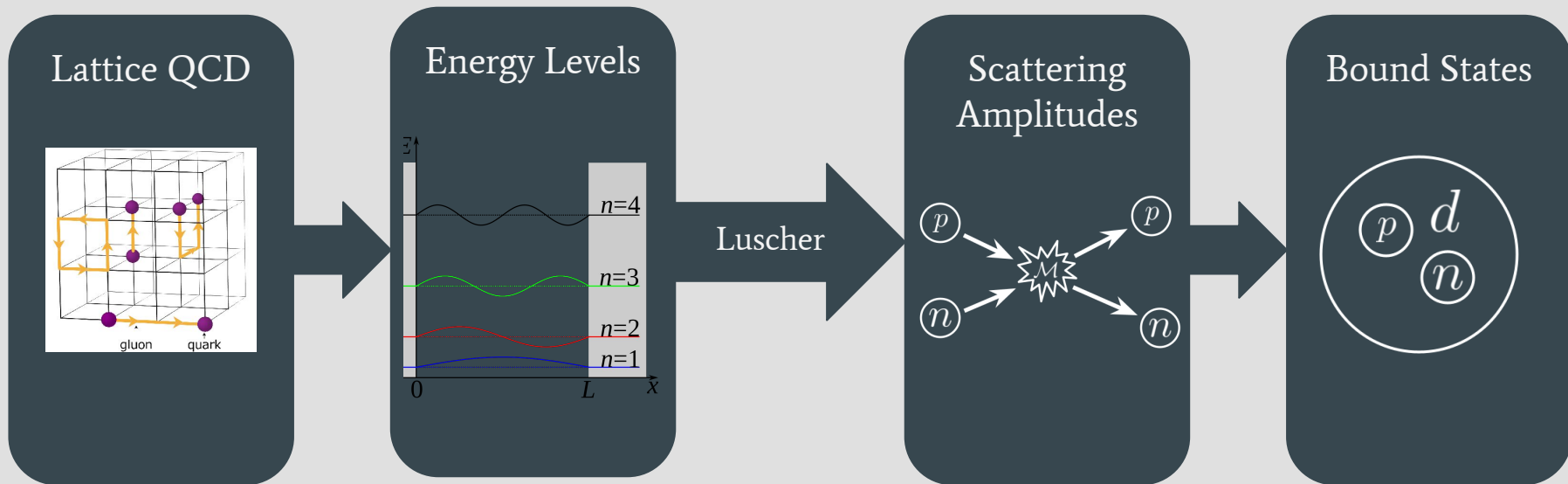
$$f_\ell \sim \Gamma\left(\ell + 1 + i\frac{\mu\alpha}{k}\right)$$

$$E_n = \frac{k^2}{2\mu} = -\frac{\mu\alpha^2}{2n^2}$$



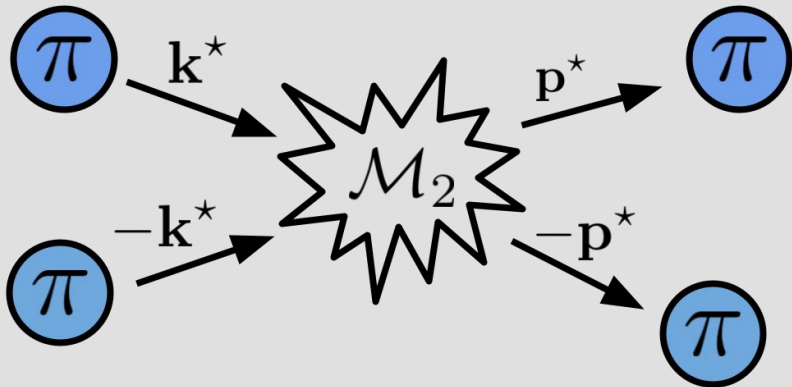
Lattice QCD

- Numerical method to compute QCD observables
- Computes energy levels of particles in a box
- Luscher method connects energy levels to scattering processes



Two-Pion Scattering

- Pions: simplest hadrons, mesons (spin-0)
- Relativistic scattering amplitude can be written for definite angular momentum
- Obtained through unitarity
- Phase shift: strength and range of interaction, depends on angular momentum
- Computations done in natural units, COM frame
- Unphysically heavy pion masses (700 MeV vs 140 MeV)

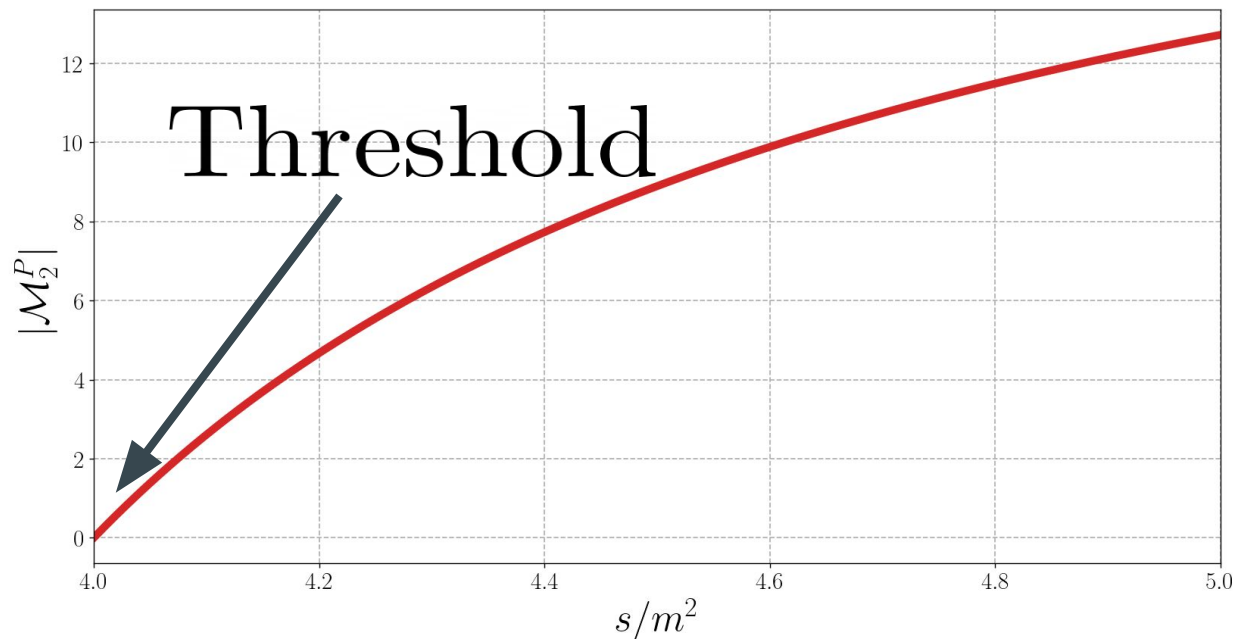


$$\mathcal{M}_2^\ell = \frac{16\pi E}{k \cot \delta_\ell - ik}$$

Two-Pion Scattering for the P-Wave Case

$$k \cot \delta_P = \frac{m_{BW}^2 - s}{\Gamma(s)\sqrt{s}}$$

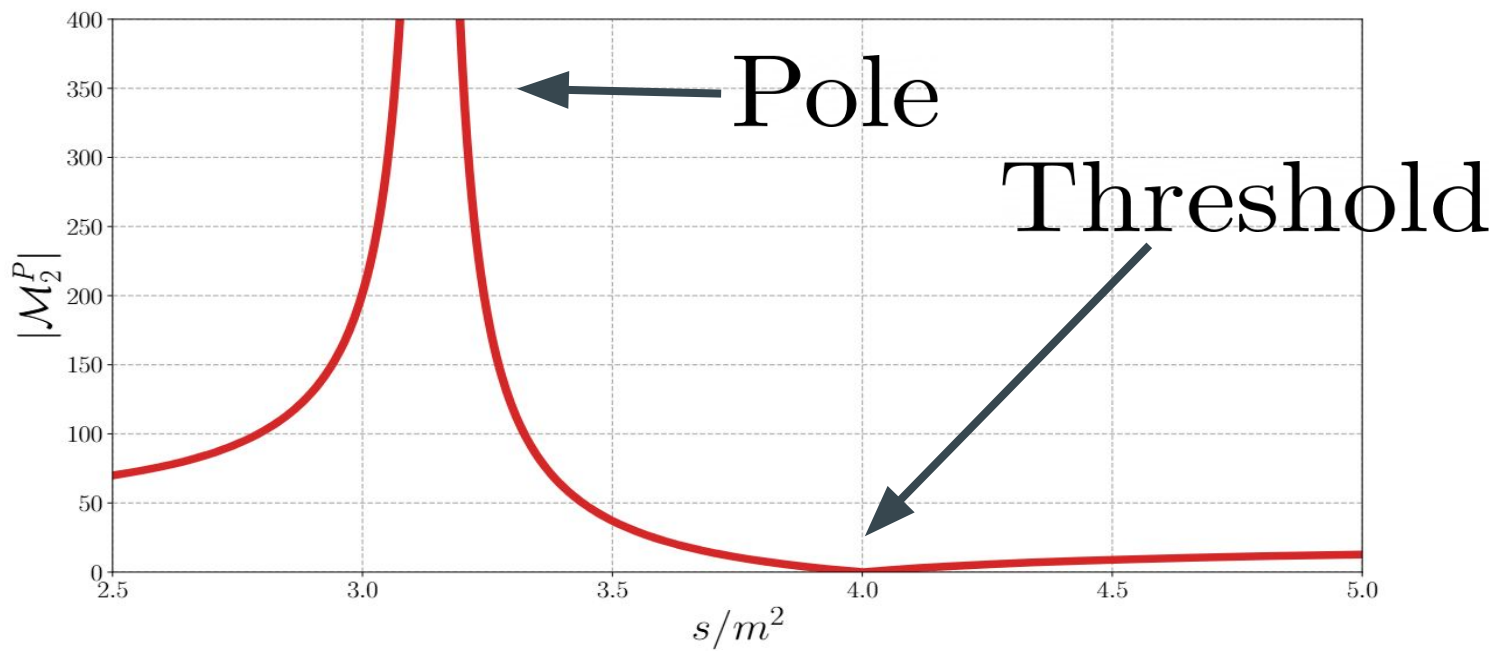
- P-wave: system where total angular momentum is 1
- Breit-Wigner parametrization, parameters come from LQCD



$$\mathcal{M}_2^\ell = \frac{16\pi E}{k \cot \delta_\ell - ik}$$

Searching for Poles

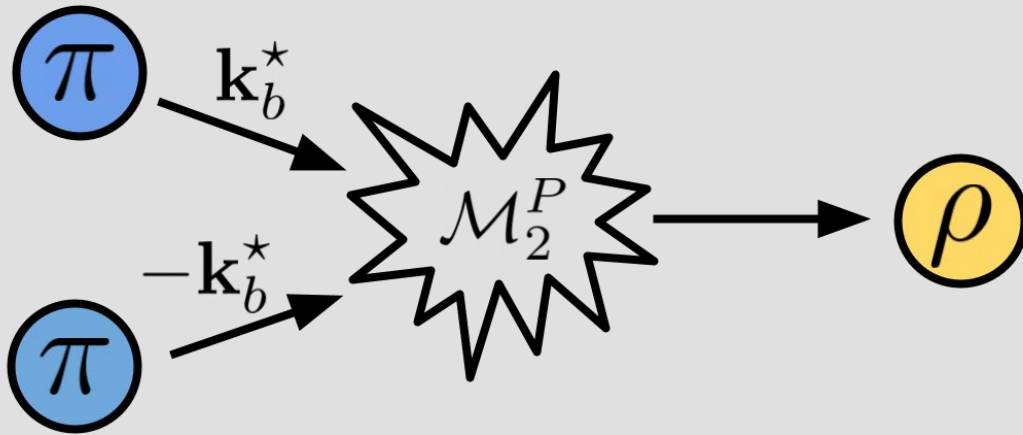
- Threshold: minimum physical energy of system, masses of particles, no momentum
- Examine scattering amplitude below threshold



The Rho Meson

$$I(J^P) = 1(1^-)$$

- Rho meson: bound state of P-wave two-pion scattering
- At a certain momentum/energy, a rho is produced instead of typical scattering



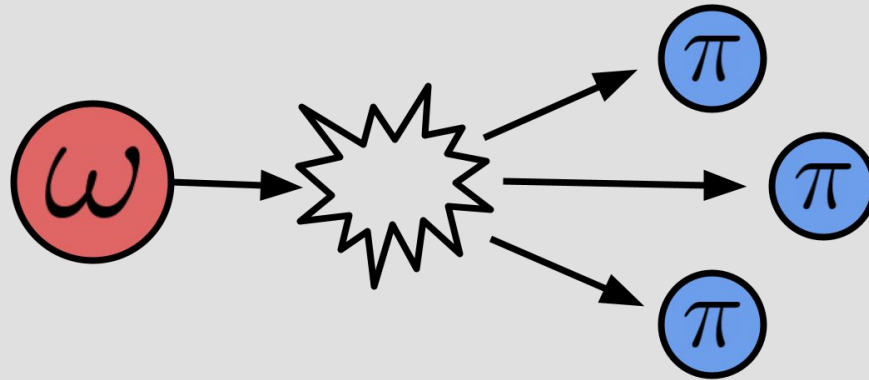
$$m_\rho \approx 1.772m$$

$$g_\rho \approx 4.886im$$

The Omega Meson

$$I(J^P) = 0(1^-)$$

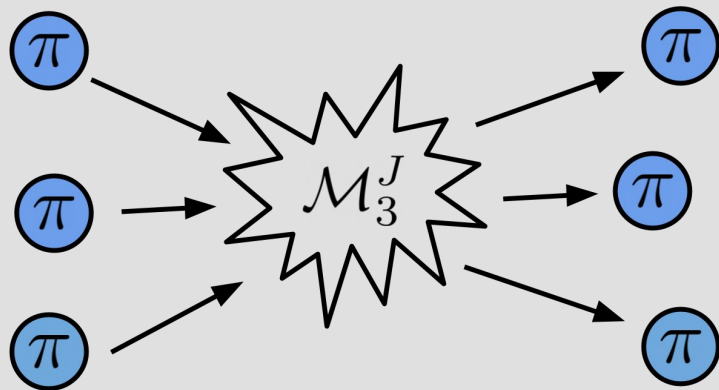
- Simplest hadron that decays into three other hadrons (pions)
- Can be studied as a product of three-pion scattering
- Three-body scattering difficult to do experimentally, must be done theoretically
- Spin 1, minus parity, and isospin 0



Three-Pion Scattering

- Similarly described by a scattering amplitude
- Three-body amplitude depends on two main interaction types:
 - Purely three-body interactions: all pions interact at once
 - Ladder interactions: pions only interact pairwise
- We want to know which of these two is the dominant mechanism to produce the omega

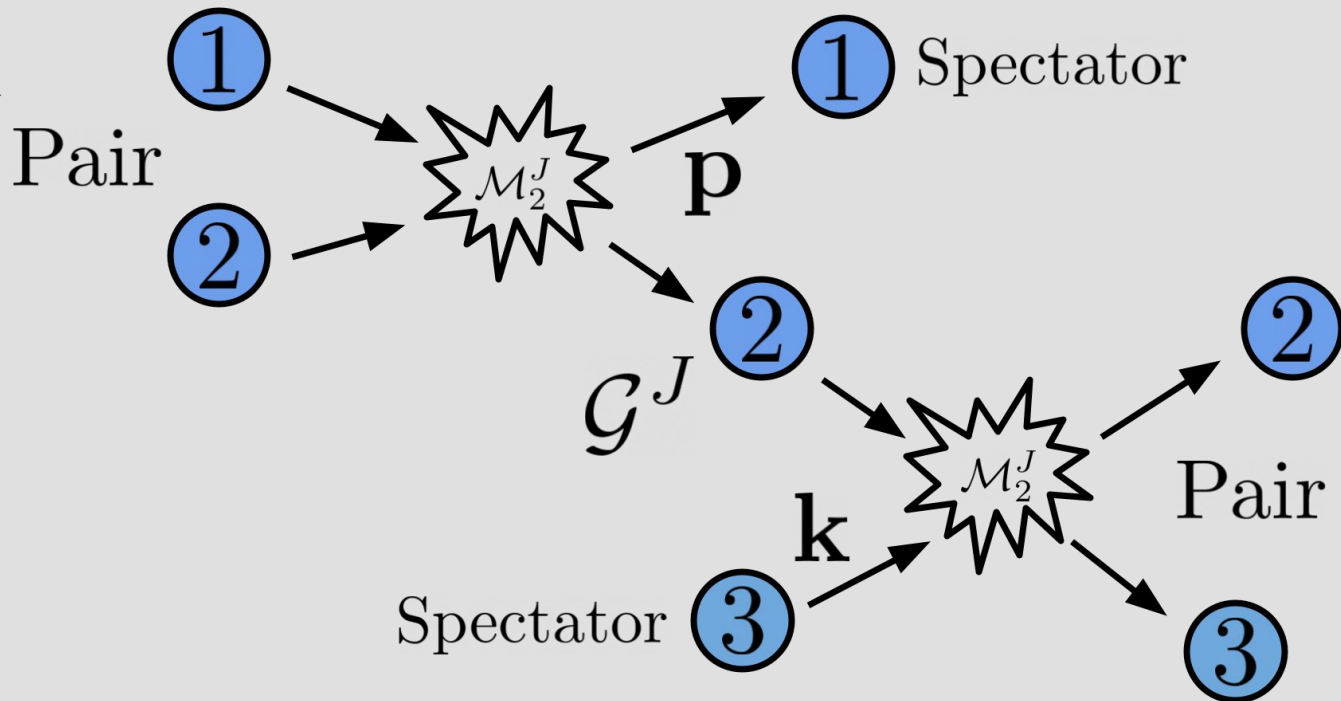
$$\mathcal{M}_3 = \mathcal{M}_{\text{df},3} + \mathcal{D}$$



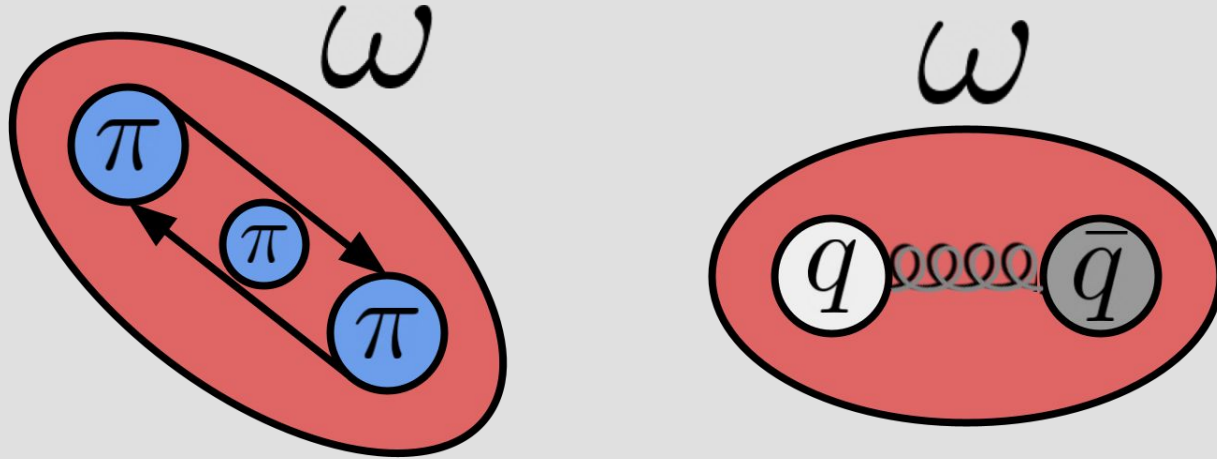
The Ladder Amplitude, $\mathcal{D}(p, k)$

- Only pairwise interactions between pions
- Depends on two functions:
 - Two-pion scattering amplitude
 - Propagator
- **Hypothesis:** ladder interactions dominate

$$\mathcal{M}_3 \approx \mathcal{D}$$



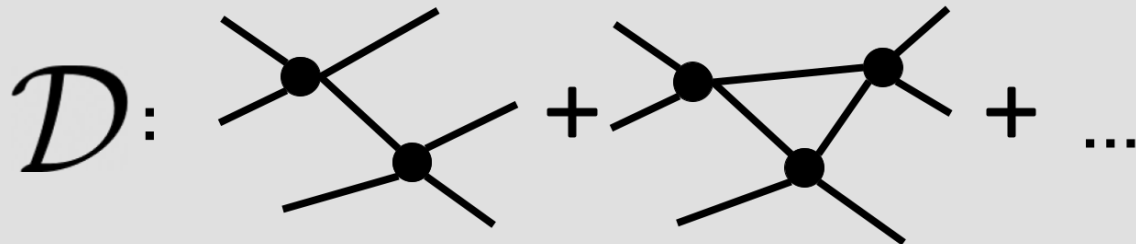
Two Possible Cases



- Older study (Kamal, 1965) supports second case
- We revisit with modern scattering theory

Ladder Amplitude Integral Equation

- Ladder amplitude described by a Fredholm integral equation of the second kind
- Cannot be solved analytically, must be done numerically
- Depends on both the two-pion scattering amplitude and the propagator

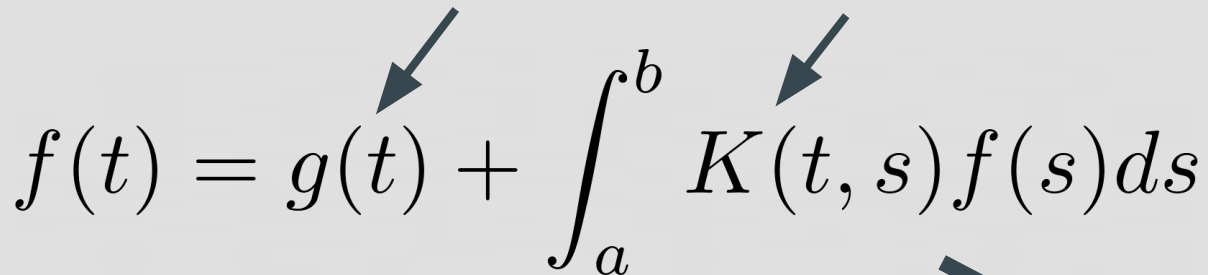


$$\mathcal{D}(p, k) = \mathcal{M}_2(\varsigma_p) \mathcal{G}(p, k) \mathcal{M}_2(\varsigma_k) - \mathcal{M}_2(\varsigma_p) \int dk' \tau(k') \mathcal{G}(p, k') \mathcal{D}(k', k)$$

Numerically Solving Integral Equations

- Functions are discretized as vectors, kernel is discretized as a matrix
- Turns equation into a linear algebra problem
- Vector solution obtained by computer, then interpolated

Driving Term Kernel


$$f(t) = g(t) + \int_a^b K(t, s) f(s) ds$$

$$\mathbf{f} = \mathbf{g} + \mathbf{M}\mathbf{f}$$

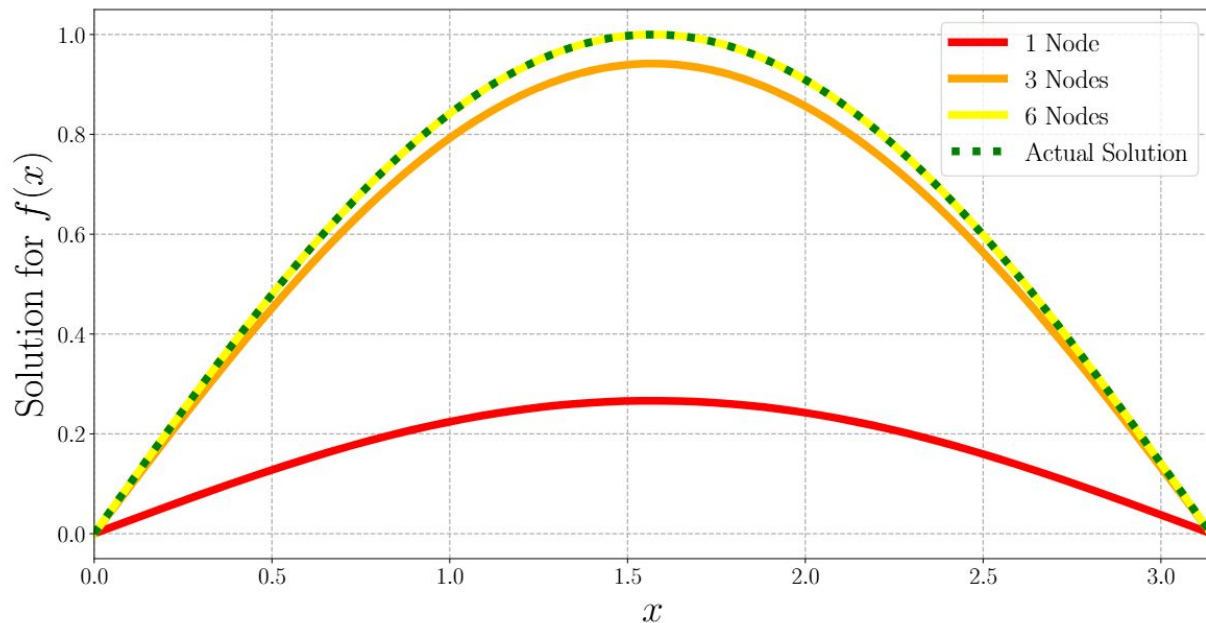
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$$\begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \end{bmatrix} = \begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_N \end{bmatrix} + \begin{bmatrix} w_1 J_1 K_{11} & w_2 J_2 K_{12} & \dots & w_N J_N K_{1N} \\ w_1 J_1 K_{21} & w_2 J_2 K_{22} & \dots & w_N J_N K_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ w_1 J_1 K_{N1} & w_2 J_2 K_{N2} & \dots & w_N J_N K_{NN} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \end{bmatrix}$$

Example Integral Equation

Solution: $f(t) = \sin(t)$

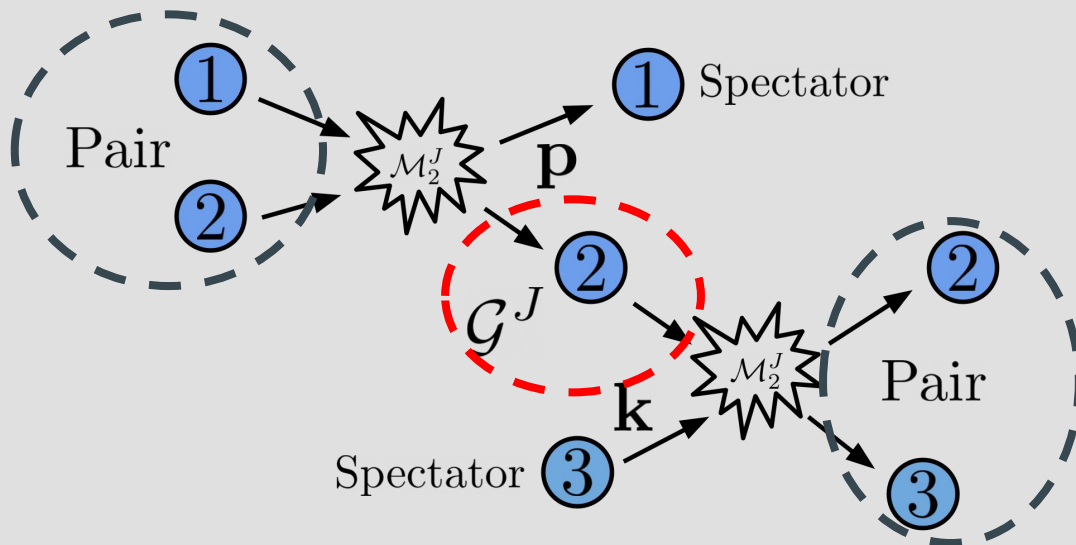


$$f(t) = \left(1 - \frac{\pi}{2}\right) \sin(t) + \int_0^{\pi} \cos(t-s) f(s) ds$$

The OPE Propagator

$$\mathcal{G} \sim \frac{1}{Q^2 - m^2}$$

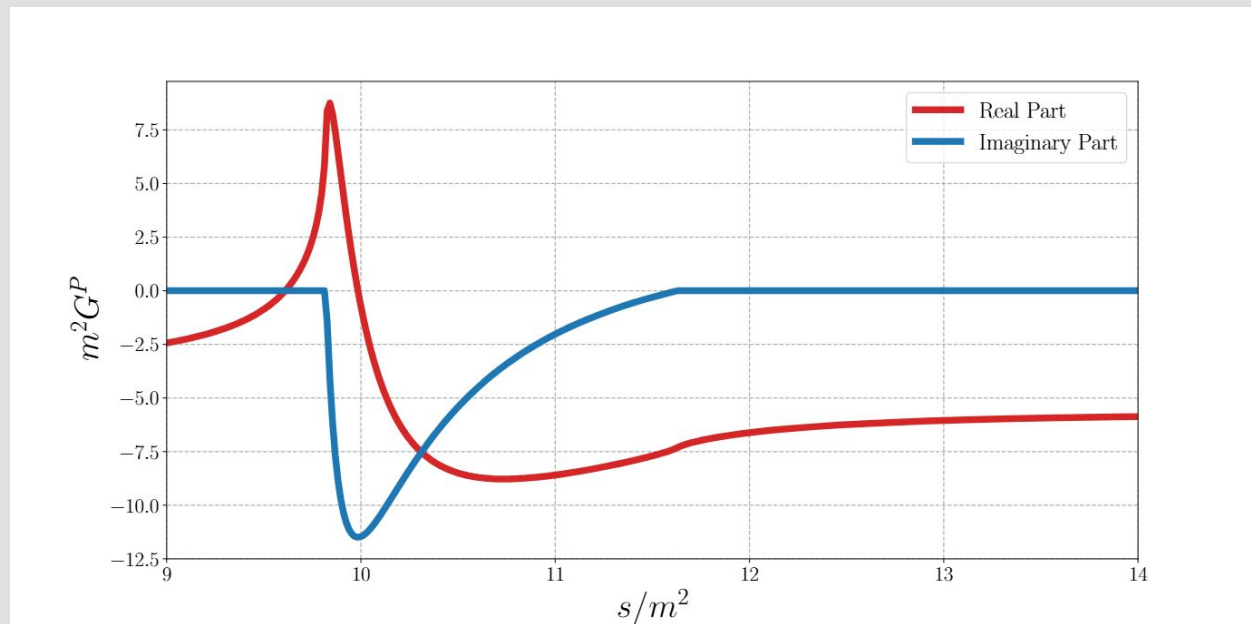
- Propagator: governs exchange of pion from one pair to another
- Poles appear when momentum transferred between the pairs is equal to pion mass



P-Wave Propagator

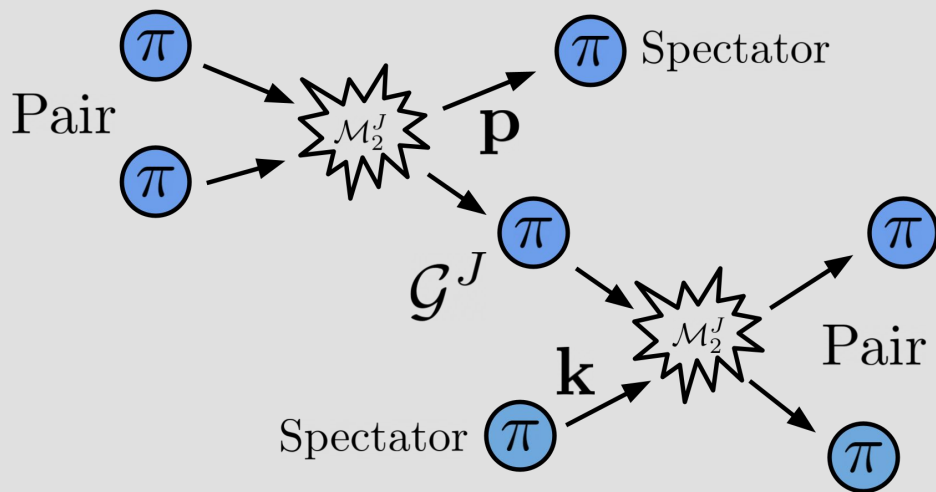
$$\mathcal{G}^P \sim \ln \left(\frac{\zeta + 1}{\zeta - 1} \right)$$

- Since ω is spin-1, we are only interested in the P-wave term of the propagator
- Angular dependence removed by integrating over solid angle, turns pole into logarithm



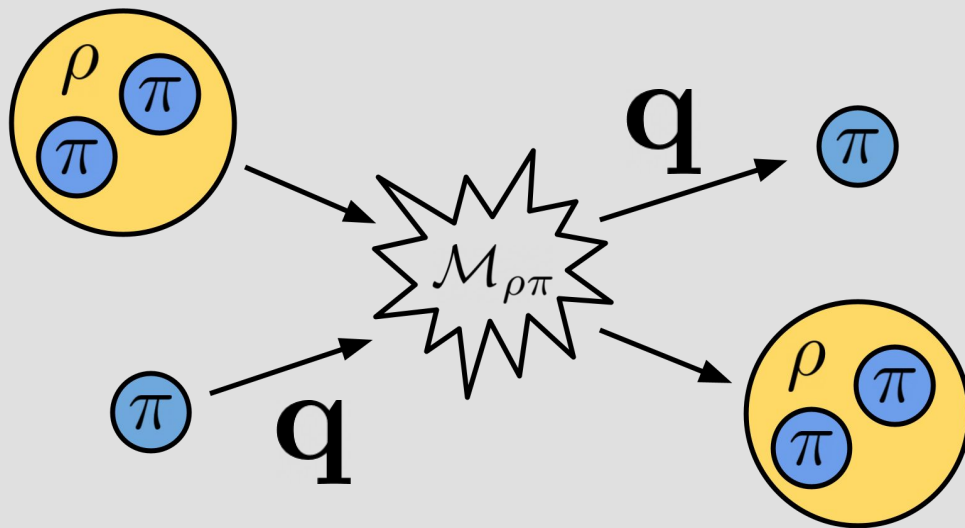
Effective “2+1” Body System

- To “build” an omega through ladder interactions, the pair must form a bound state rather than scatter ($\pi\pi \rightarrow \rho$)
- Solve for spectator momentum that produces 2-body pole, denoted q
- Access this subprocess using the LSZ theorem (Gribov, 1968), find ladder amplitude near poles of 2-body amplitude



Effective “2+1” Body System

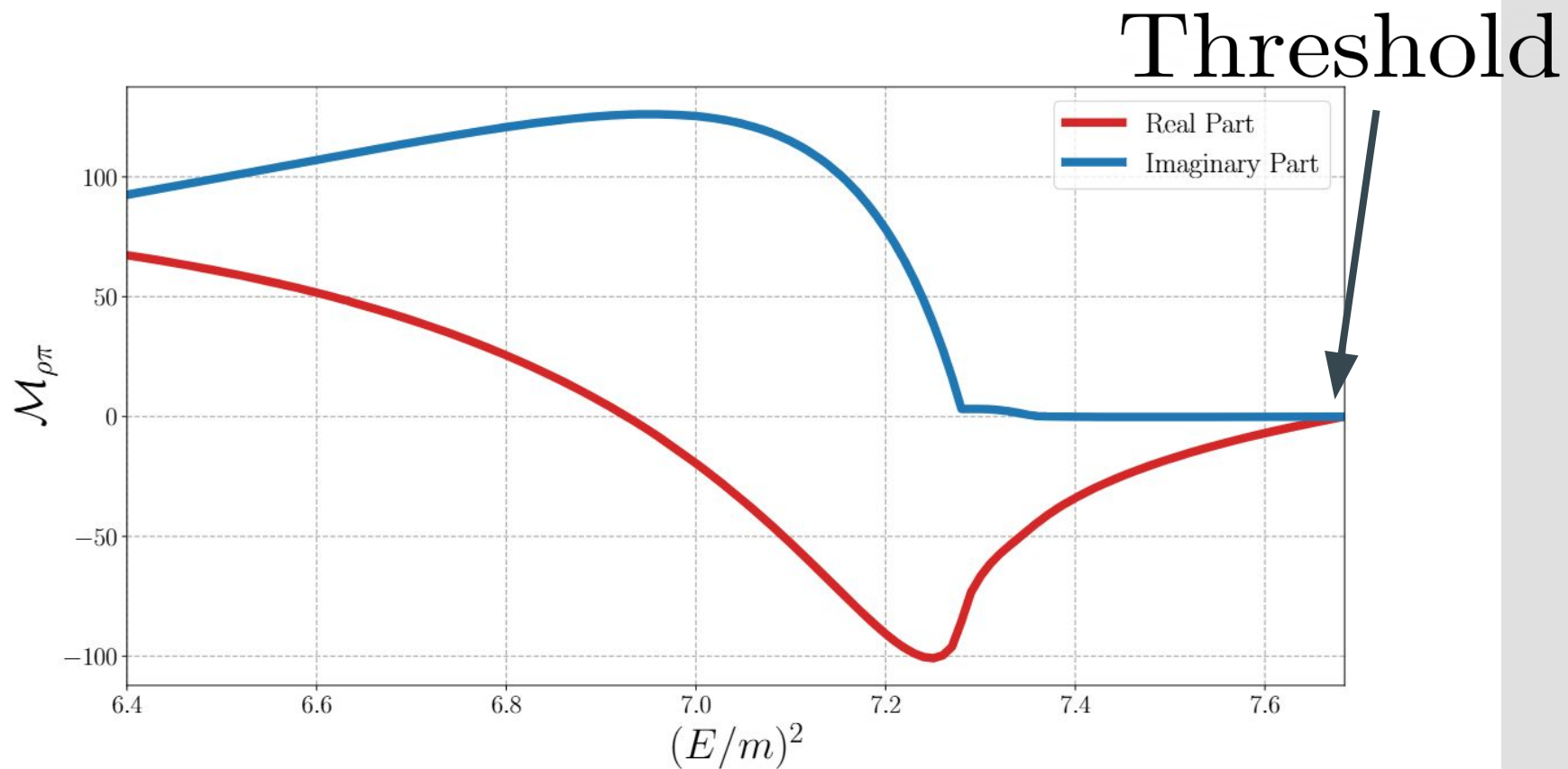
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$$\mathcal{M}_{\rho\pi}(s) = g_\rho^2 d^P(q, q)$$

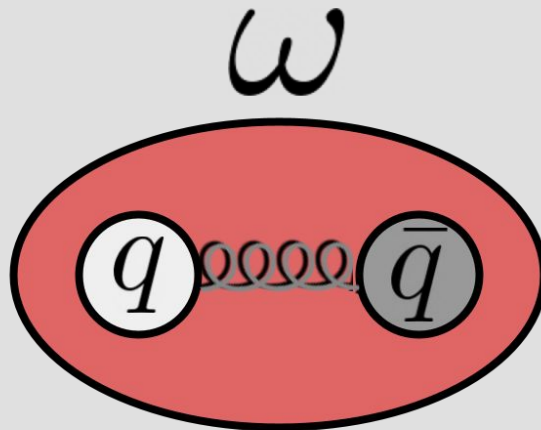
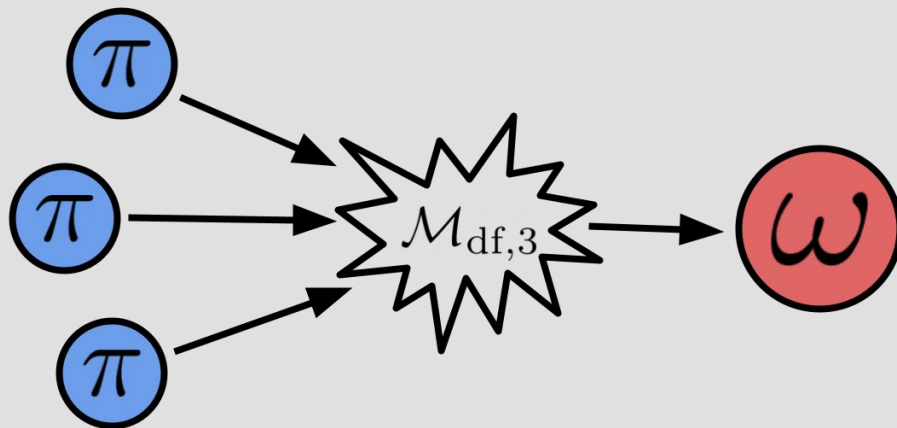
$$\mathcal{D}^P = \mathcal{M}_2(\varsigma_k) d^P \mathcal{M}_2(\varsigma_p)$$

Solution Below Threshold



Conclusions

- No pole found - **hypothesis is false**
- Purely three-body forces must play significant role
- Results align with (Kamal, 1965)
- Next steps: purely three-body interactions must be studied through lattice QCD



References

- [1] S. Navas et al. “Review of particle physics”. In: *Phys. Rev. D* 110.3 (2024), p. 030001. DOI: [10.1103/PhysRevD.110.030001](https://doi.org/10.1103/PhysRevD.110.030001).
- [2] Vladimir Gribov. *Strong Interactions of Hadrons at High Energies*. Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology. Prepared by Y. Dokshitzer and J. Nyiri. Cambridge: Cambridge University Press, 1997.
- [3] Raul A. Briceño et al. “Isoscalar $\pi\pi$ Scattering and the σ Meson Resonance from QCD”. In: *Phys. Rev. Lett.* 118 (2 Jan. 2017), p. 022002. DOI: [10.1103/PhysRevLett.118.022002](https://doi.org/10.1103/PhysRevLett.118.022002). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.118.022002>.
- [4] Thomas Jefferson National Accelerator Facility. *Jefferson Lab Official Website*. Accessed: 2025-03-28. 2025. URL: <https://www.jlab.org>.
- [5] Walter Gloeckle, T. Cornelius, and H. Witala. “Three nucleon scattering: A test for nuclear dynamics”. In: *Nucl. Phys. A* 508 (1990). Ed. by H. W. Fearing, pp. 115C–130C. DOI: [10.1016/0375-9474\(90\)90468-2](https://doi.org/10.1016/0375-9474(90)90468-2).
- [6] M. Viviani et al. “Effect of Three Nucleon Forces in $p-^3\text{He}$ Scattering”. In: *Few Body Syst.* 54 (2013). Ed. by K. Sagara et al., pp. 885–890. DOI: [10.1007/s00601-013-0665-6](https://doi.org/10.1007/s00601-013-0665-6). arXiv: [1210.5890](https://arxiv.org/abs/1210.5890) [nucl-th].
- [7] M. Viviani et al. “Three-nucleon force effects in $A = 4$ scattering”. In: *Few Body Syst.* 59.4 (2018), p. 73. DOI: [10.1007/s00601-018-1379-6](https://doi.org/10.1007/s00601-018-1379-6).
- [8] A. N. Kamal. “The role of OPE mechanism in binding the ω -meson”. In: *Nuovo Cim. A* 63.1 (1965), pp. 126–141. DOI: [10.1007/BF02898814](https://doi.org/10.1007/BF02898814).
- [9] Andrew W. Jackura et al. “Solving relativistic three-body integral equations in the presence of bound states”. In: *Phys. Rev. D* 104.1 (2021), p. 014507. DOI: [10.1103/PhysRevD.104.014507](https://doi.org/10.1103/PhysRevD.104.014507). arXiv: [2010.09820](https://arxiv.org/abs/2010.09820) [hep-lat].
- [10] Andrew W. Jackura and Raúl A. Briceño. “Partial-wave projection of the one-particle exchange in three-body scattering amplitudes”. In: (Dec. 2023). arXiv: [2312.00625](https://arxiv.org/abs/2312.00625) [hep-ph].
- [11] Jozef J. Dudek et al. “Toward the excited meson spectrum of dynamical QCD”. In: *Phys. Rev. D* 82 (2010), p. 034508. DOI: [10.1103/PhysRevD.82.034508](https://doi.org/10.1103/PhysRevD.82.034508). arXiv: [1004.4930](https://arxiv.org/abs/1004.4930) [hep-ph].
- [12] Jozef J. Dudek, Robert G. Edwards, and Christopher E. Thomas. “Energy dependence of the ρ resonance in $\pi\pi$ elastic scattering from lattice QCD”. In: *Phys. Rev. D* 87.3 (2013). [Erratum: *Phys. Rev. D* 90, 099902 (2014)], p. 034505. DOI: [10.1103/PhysRevD.87.034505](https://doi.org/10.1103/PhysRevD.87.034505). arXiv: [1212.0830](https://arxiv.org/abs/1212.0830) [hep-ph].
- [13] Maxwell T. Hansen and Stephen R. Sharpe. “Relativistic, model-independent, three-particle quantization condition”. In: *Phys. Rev. D* 90.11 (2014), p. 116003. DOI: [10.1103/PhysRevD.90.116003](https://doi.org/10.1103/PhysRevD.90.116003). arXiv: [1408.5933](https://arxiv.org/abs/1408.5933) [hep-lat].
- [14] Maxwell T. Hansen and Stephen R. Sharpe. “Expressing the three-particle finite-volume spectrum in terms of the three-to-three scattering amplitude”. In: *Phys. Rev. D* 92.11 (2015), p. 114509. DOI: [10.1103/PhysRevD.92.114509](https://doi.org/10.1103/PhysRevD.92.114509). arXiv: [1504.04248](https://arxiv.org/abs/1504.04248) [hep-lat].
- [15] Andrew W. Jackura. “Three-body scattering and quantization conditions from S-matrix unitarity”. In: *Phys. Rev. D* 108.3 (2023), p. 034505. DOI: [10.1103/PhysRevD.108.034505](https://doi.org/10.1103/PhysRevD.108.034505). arXiv: [2208.10587](https://arxiv.org/abs/2208.10587) [hep-lat].
- [16] John S. Townsend. *A Modern Approach to Quantum Mechanics*. 2nd. Sausalito, CA: University Science Books, 2012.
- [17] Jun John Sakurai. *Modern quantum mechanics; rev. ed.* Reading, MA: Addison-Wesley, 1994. URL: <https://cds.cern.ch/record/1167961>.
- [18] V. N. Gribov. *Quantum Electrodynamics: Gribov Lectures on Theoretical Physics*. Vol. 13. Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology. Cambridge University Press, 2001. ISBN: 9780521630647. URL: <https://doi.org/10.1017/CB09780511524370>.
- [19] Raúl A. Briceño, Caroline S. R. Costa, and Andrew W. Jackura. “Partial-wave projection of relativistic three-body amplitudes”. In: (2024). Dated: September 25, 2024.
- [20] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to Quantum Field Theory*. Boulder, CO: Westview Press, 1995. ISBN: 978-0-201-50397-5.
- [21] L. M. Delves and J. L. Mohamed. *Computational Methods for Integral Equations*. Cambridge: Cambridge University Press, 1985. ISBN: 9780521308477.
- [22] Sebastian M. Dawid, Md Habib E. Islam, and Raúl A. Briceño. “Analytic continuation of the relativistic three-particle scattering amplitudes”. In: *Phys. Rev. D* 108.3 (2023), p. 034016. DOI: [10.1103/PhysRevD.108.034016](https://doi.org/10.1103/PhysRevD.108.034016). arXiv: [2303.04394](https://arxiv.org/abs/2303.04394) [nucl-th].
- [23] Raúl A. Briceño, Caroline S. R. Costa, and Andrew W. Jackura. “Partial-wave projection of relativistic three-body amplitudes”. In: *Phys. Rev. D* 111 (3 Feb. 2025), p. 036029. DOI: [10.1103/PhysRevD.111.036029](https://doi.org/10.1103/PhysRevD.111.036029). URL: <https://link.aps.org/doi/10.1103/PhysRevD.111.036029>.

Questions?

Unitarity

- Probability must be conserved, places the following constraint on scattering amplitude

$$\Im[\mathcal{M}_2^\ell] = \varrho |\mathcal{M}_2^\ell|^2$$

$$\mathcal{M}_2^{-1} = \mathcal{K}_2^{-1} - i\varrho$$

$$\mathcal{M}_2^\ell = \frac{16\pi E}{k \cot \delta_\ell - ik}$$

Full 2-Body Amplitude

$$\mathcal{M}_2^\ell = \frac{16\pi E}{k \cot \delta_\ell - ik} \quad k \cot \delta_P = \frac{m_{BW}^2 - s}{\Gamma(s)\sqrt{s}}$$

$$\Gamma_1^{\text{BW}}(s) = \frac{g_{\text{BW}}^2}{6\pi s} k^{\star 2} \quad m_{BW} = 1.8m$$
$$g_{BW} = 5.8$$

Full Propagator

$$\mathcal{G}^P(p, k) = -\frac{3}{4q_p^\star q_k^\star} \zeta_{pk} + \frac{3}{4q_p^\star q_k^\star} (\zeta_{pk} - 1) Q_0(\zeta_{pk})$$

$$Q_0(x) = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right)$$

$$q_k^\star = \sqrt{\varsigma_k/4 - m^2}$$

$$\zeta_{pk} := \frac{2E^2 \varsigma_k - (E^2 + \varsigma_k - m^2)(E^2 + m^2 - \varsigma_p)}{\lambda^{1/2}(E^2, \varsigma_k, m^2) \lambda^{1/2}(E^2, \varsigma_p, m^2)}$$

$$\lambda(x, y, z) := x^2 + y^2 + z^2 - 2(xy + yz + zx)$$

Partial Wave Projection

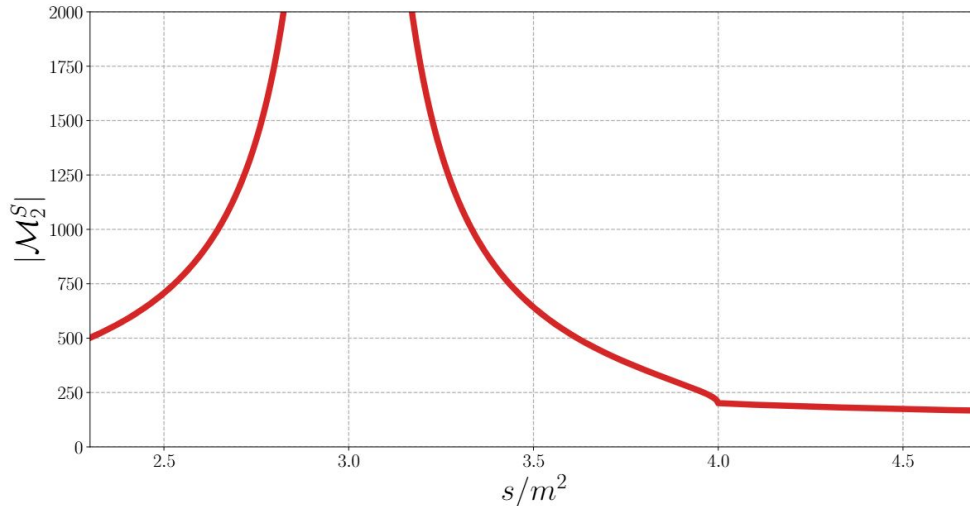
- Split scattering amplitude up into sum of definite angular momentum terms

$$\mathcal{M}_2(E^*, \theta^*) = \sum_{\ell} (2\ell + 1) \mathcal{M}_2^{\ell}(E^*) P_{\ell}(\cos(\theta^*))$$

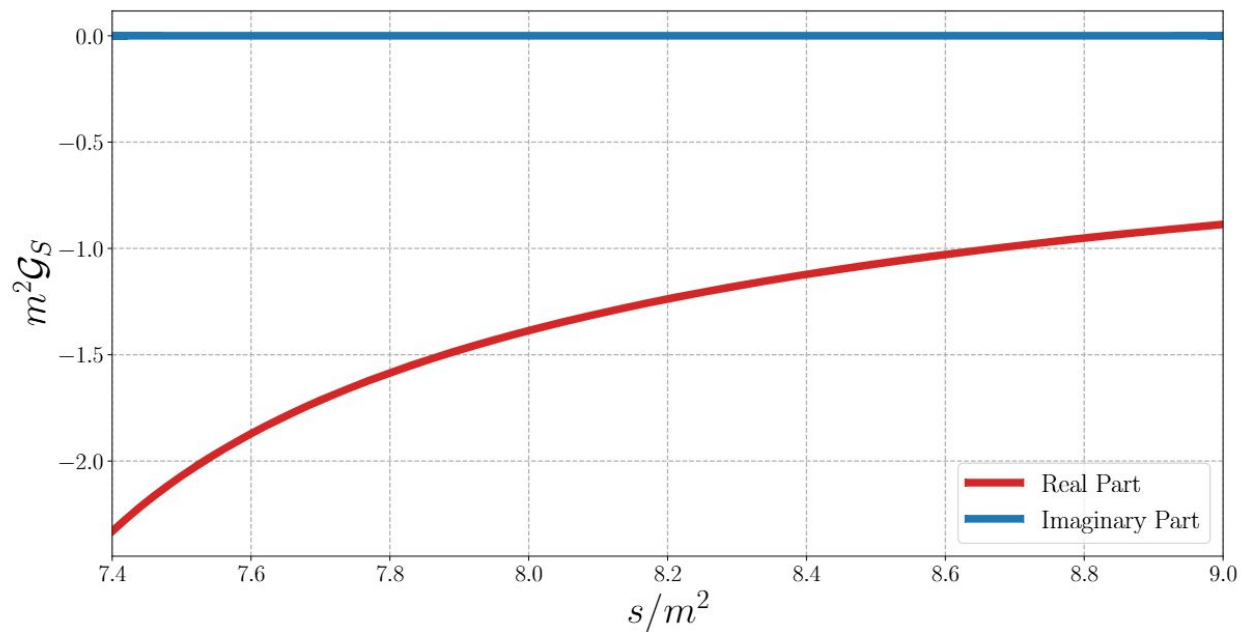
The Sigma Meson

$$I(J^P) = 0(0^+)$$

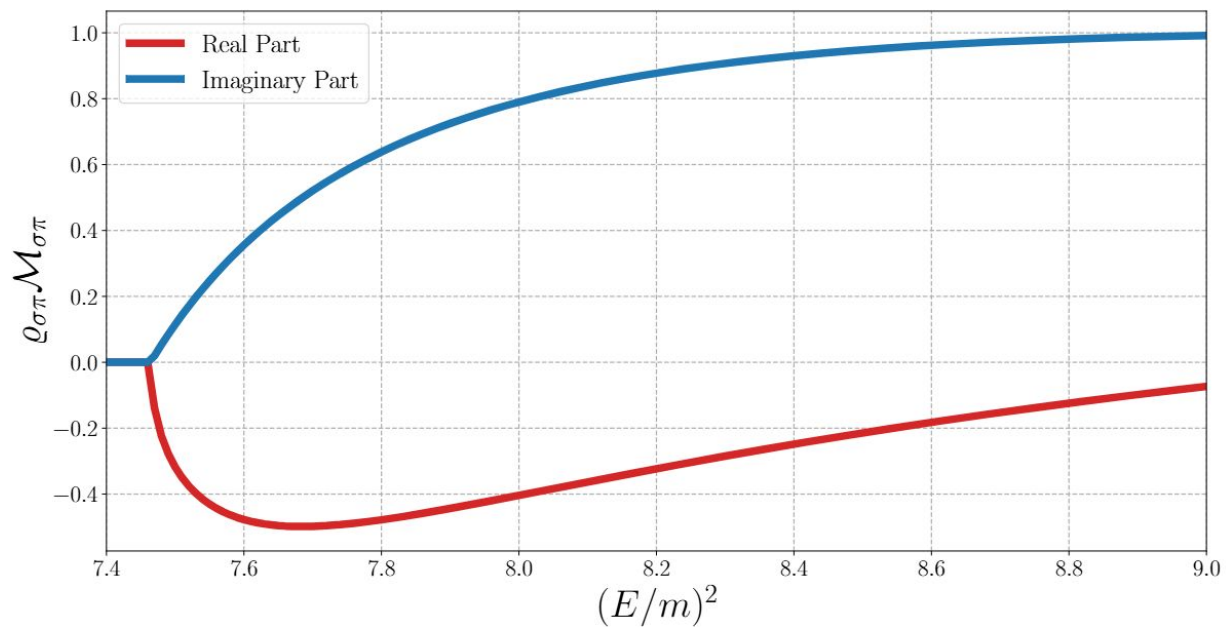
- Spin-0 resonance from 2-pion scattering
- Found outside the quark model
- Was used as a “test case” in this thesis, as spin 0 amplitudes and propagators are less complicated



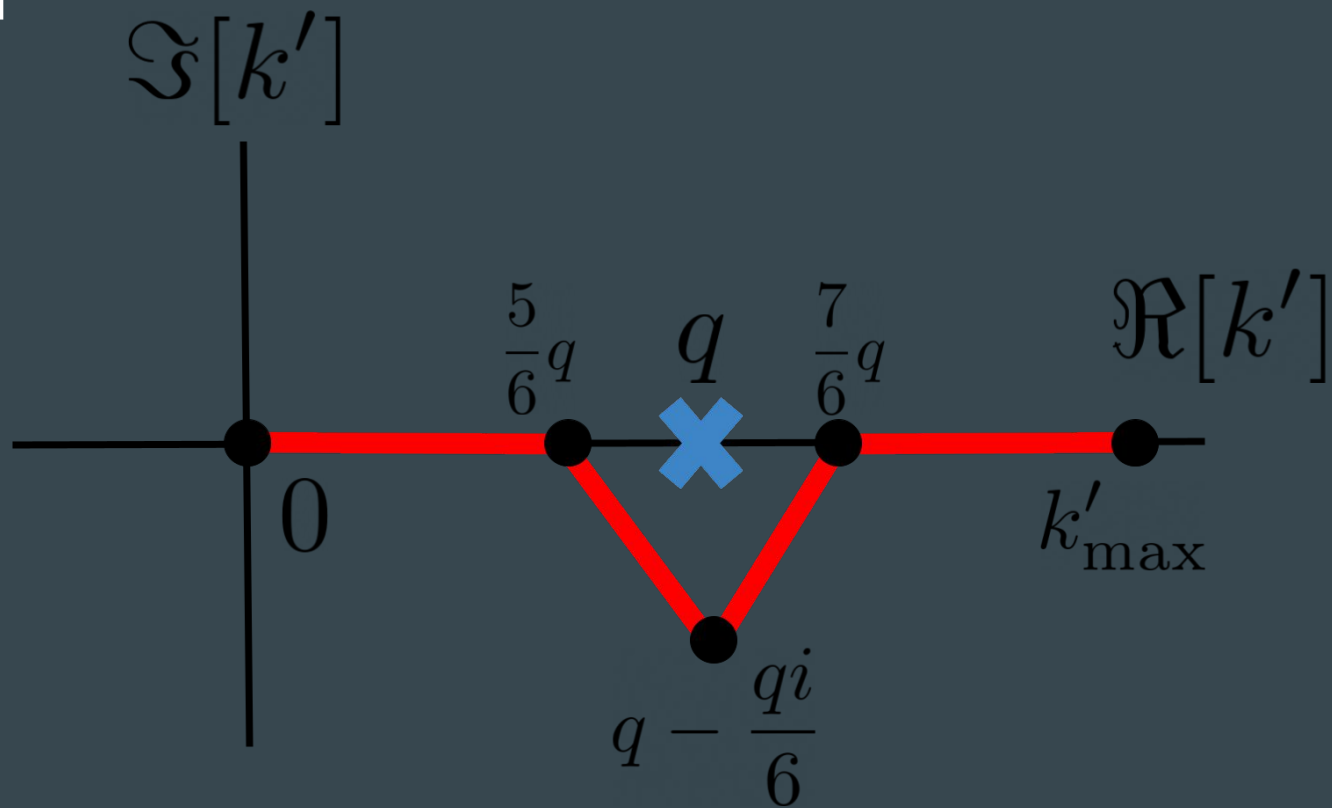
S-Wave Propagator



Sigma-Pi Amplitude



Integration Path



Other Three-Body Hadronic Reactions

- Stellar nucleosynthesis

